PHYSIOLOGY

Physiology of Urine Formation

SGL 1
Dr. Dler Qader Gallaly
GUE BLOCK

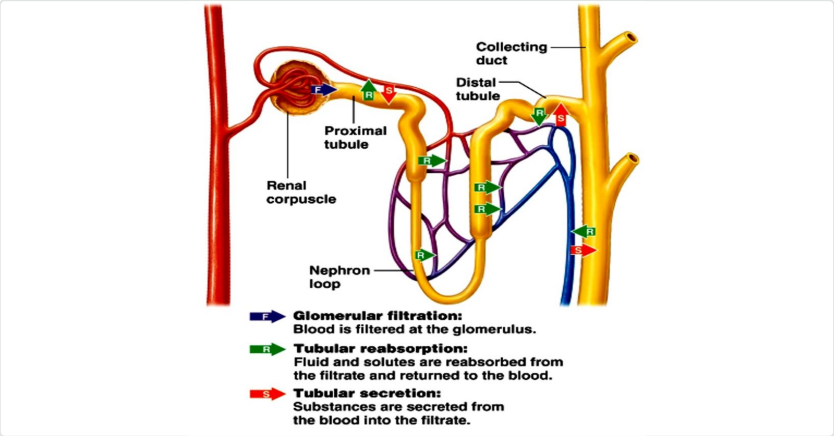
Physiology GUE Block · MEDucation by Mattin Majid

BEFORE YOU READ — ANSWER FROM MEMORY

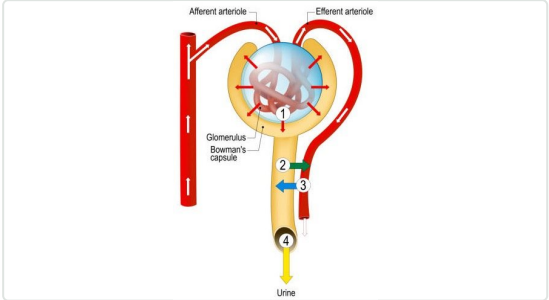
1. What are the three processes of urine formation, in order?
2. What is the normal GFR, and which three pressures determine net filtration pressure (and their values)?
3. Which substances are 100% reabsorbed in the PCT, and by what transport mechanism?

Guessing wrong before reading still improves retention — attempt all three, then check as you go.

1 The three-process concept map



Blood is **filtered** at the glomerulus (F); solutes are **reabsorbed** back to blood all along the tubule (R); unwanted solutes are **secreted** into the tubule (S).



① Glomerular filtration → ② Reabsorption → ③ Secretion → ④ Urine leaves via the collecting duct.

Process	Direction	What happens
1. Glomerular filtration	Blood → Bowman's capsule	Plasma filtered across glomerular capillaries; filtrate lacks cells/large proteins
2. Tubular reabsorption	Tubule → peritubular blood	**Wanted** substances (glucose, amino acids, water, electrolytes) returned to blood — "selective reabsorption"
3. Tubular secretion	Peritubular blood → tubule	**Unwanted** substances added into the filtrate for excretion

Urine formation = the kidneys' **blood-cleansing function**. ~1300 mL/min of blood enters the kidneys (26% of cardiac output); normal urinary output is **1–1.5 L/day**.

2 Glomerular filtration rate (GFR)

Term	Definition / value
Ultrafiltration	Glomerular filtration filters even minute particles, but **plasma proteins are not filtered** — too large for the capillary endothelium's filtration pores
GFR	Total filtrate formed by **both kidneys per unit time** — normal ≈ **125 mL/min = 180 L/day**
Filtration fraction	The **ratio of renal plasma flow to GFR** — the fraction of renal plasma that actually becomes filtrate

3 Net filtration pressure (NFP) — 3 Starling forces

Pressure	Value	Effect on filtration
Glomerular capillary pressure	≈60 mmHg (range 45–70) — highest capillary pressure in the body	**Favors**
Colloidal osmotic (oncotic) pressure	≈25 mmHg — from plasma proteins trapped in the capillary	**Opposes**

Hydrostatic (capsular) pressure, Bowman's capsule	≈15 mmHg — exerted by the filtrate itself	**Opposes**

NET FILTRATION PRESSURE

NFP = Glomerular capillary pressure − (Colloidal osmotic pressure + Capsular pressure) = 60 − (25 + 15) = 20 mmHg (varies 15–20 mmHg)

Filtration coefficient (FC) = GFR ÷ NFP = 125 mL/min ÷ 20 mmHg = 6.25 mL/min/mmHg

4 Factors affecting GFR

Renal blood flow (most important — GFR directly proportional) · Tubuloglomerular feedback (macula densa senses NaCl, adjusts GFR) · Glomerular capillary pressure · Colloidal osmotic pressure · Bowman's capsule pressure · Afferent arteriole constriction (↓GFR) · Efferent arteriole constriction (↑GFR initially) · Systemic arterial pressure · Sympathetic stimulation · Capillary surface area · Capillary permeability · Hormonal factors

5 Tubular reabsorption — routes & key substances

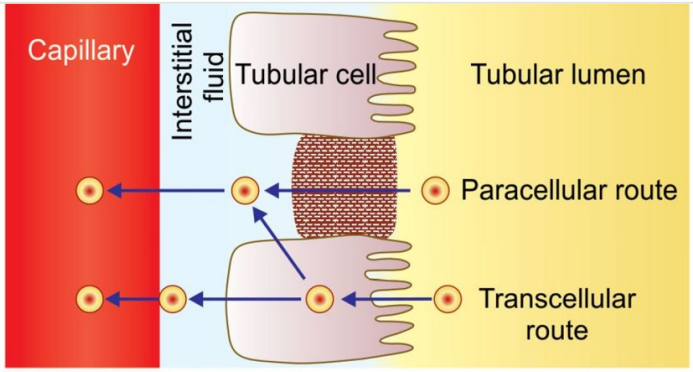


FIGURE 50.3: Routes of reabsorption.

Two routes from tubular lumen to peritubular capillary.

Route	Path
Transcellular	Lumen → **through** the tubular cell (apical surface → cell → interstitial fluid → capillary)
Paracellular	Lumen → through the **tight junction** between cells → interstitial fluid → capillary

Segment	Substances reabsorbed
Proximal convoluted tubule (PCT)	Glucose, amino acids, Na^+, K^+, Ca^{2+}, HCO_3^-, Cl^-, phosphate, urea, uric acid, water
Loop of Henle	Na^+, Cl^-
Distal convoluted tubule (DCT)	Na^+, Ca^{2+}, Cl^-, HCO_3^-, water
Collecting duct (CD)	Na^+ (aldosterone-dependent)

Substance	Key fact
Water	>99% reabsorbed, from PCT, DCT and CD
Sodium	99% reabsorbed; 2/3 (66%) reabsorbed in PCT + CD together
Glucose & amino acids	**Completely** reabsorbed in the PCT, by secondary active transport (Na^+-cotransport)
Bicarbonate	Reabsorbed actively, mostly in the proximal tubule, **in the form of CO_2**

Tubular transport maximum (Tm): for every actively reabsorbed substance there is a maximum rate at which it can be reabsorbed — exceed it (e.g. very high blood glucose) and the substance appears in urine.

SOURCE NOTE

The lecture gives sodium reabsorption as "2/3 (66%) in PCT and CD" combined, rather than breaking out each segment's individual share. Worth cross-checking against Dr. Sherwan's Water & Electrolyte lecture (which states PCT alone reabsorbs ≈65% of filtered Na^+) if an exam question asks for a single-segment figure.

6 Tubular secretion

Substance	Site secreted
Potassium (K^+)	PCT, DCT, CD — secreted actively by the Na^+-K^+ pump
Ammonia (NH_3)	PCT
Hydrogen ions (H^+)	PCT, DCT
Urea	Loop of Henle

DON'T MIX THESE UP

Reabsorption = tubule → blood (lumen loses the substance). **Secretion** = blood → tubule (lumen gains the substance). Opposite directions — a classic exam trap.

P Must-remember pearls

- GFR ≈ **125 mL/min = 180 L/day**; only 1–1.5 L/day leaves as urine — **>99% of filtrate is reabsorbed**.
- **NFP = Glomerular capillary pressure − (Colloidal osmotic pressure + Capsular pressure) = 60 − (25+15) = 20 mmHg.**
- Glucose and amino acids are normally **100% reabsorbed in the PCT** by secondary active (Na^+-cotransport) transport — glucosuria signals the **transport maximum (Tm)** has been exceeded.
- Plasma proteins are **not filtered** — too large for the glomerular capillary pores — this is why glomerular filtration is called **ultrafiltration**.
- **Secretion** adds substances INTO the tubule (blood → lumen); **reabsorption** removes them FROM the tubule (lumen → blood) — opposite directions.

Q Test yourself

1. A student calculates net filtration pressure using a glomerular capillary pressure of 60 mmHg, a colloidal osmotic pressure of 25 mmHg, and a Bowman's capsule hydrostatic pressure of 15 mmHg. What is the correct NFP, and does it favor or oppose filtration?

A. 100 mmHg, favors filtration
B. 20 mmHg, favors filtration
C. 20 mmHg, opposes filtration
D. 40 mmHg, favors filtration
E. 10 mmHg, opposes filtration

2. A disease selectively destroys proximal tubular Na^+-glucose cotransporters (SGLT) without affecting glomerular filtration. What would you expect to find?

A. Normal urine glucose, since filtration is unaffected
B. Glucosuria despite normal blood glucose
C. Proteinuria
D. Hematuria
E. Complete absence of urine output

3. Which of the following best explains why plasma proteins normally do not appear in the glomerular filtrate?

A. They are actively secreted back into the blood
B. They are too large to pass through the glomerular filtration pores (ultrafiltration)
C. They are reabsorbed in the loop of Henle
D. They are enzymatically destroyed in the tubule
E. They bind to red blood cells and cannot filter

4. A patient develops urinary tract obstruction, raising the hydrostatic pressure inside Bowman's capsule. What is the expected effect on net filtration pressure and GFR?

A. NFP and GFR both increase
B. NFP and GFR both decrease
C. No effect on NFP
D. Colloidal osmotic pressure rises instead
E. Reabsorption converts into secretion

5. Which statement correctly distinguishes tubular reabsorption from tubular secretion?

A. Reabsorption moves substances from blood into the tubule; secretion moves them from tubule into blood
B. Both processes move substances in the same direction, blood to tubule
C. Reabsorption moves substances from tubule back to blood; secretion moves them from blood into the tubule
D. Secretion only occurs at the glomerulus
E. Reabsorption is always passive and secretion is always active

ANSWERS — COVER UNTIL YOU'VE COMMITTED

1 — B — NFP = 60 − (25+15) = 20 mmHg, and this positive value favors filtration.

2 — B — glucose is normally reabsorbed via Na^+-cotransport in the PCT; losing this transporter leaves filtered glucose in the tubule → glucosuria with normal blood glucose.

3 — B — ultrafiltration excludes plasma proteins purely by size, not by active transport or enzymatic destruction.

4 — B — Bowman's capsule hydrostatic pressure opposes filtration; raising it lowers NFP and therefore GFR.

5 — C — reabsorption is tubule→blood; secretion is blood→tubule. These are opposite directions.

SPACED REVIEW

☐ Day 1 | ☐ Day 3 | ☐ Day 7 | ☐ Day 21 | ☐ Pre-exam